

SORTLAND, Norway

WMO: 011670

Lat: 68.700N

Lon: 15.417E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.1	16.4	3.8	6.7	18.3	6.6	7.7	19.9	28.8	33.8	25.1	34.2	3.6	300	0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	68.8	57.4	64.6	56.1	61.6	55.2	59.8	64.8	58.0	62.5	56.5	60.6	8.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.8	71.5	61.1	56.1	67.0	59.5	54.3	62.8	58.0	26.3	64.8	25.1	62.7	24.1	60.8	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 23.4	20.2	17.6	DB	10.8	76.2	2.2	4.8	9.2	79.6	7.9	82.4	6.7	85.1	5.1	88.6	(4)
(5)			WB	9.5	62.7	2.1	2.5	8.0	64.5	6.7	66.0	5.5	67.5	4.0	69.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.5	30.0	29.2	31.1	36.4	43.8	49.2	54.5	53.8	48.8	41.1	35.0	32.1		
	DBStd	10.71	6.68	6.72	5.93	5.56	5.83	5.10	4.89	4.64	5.10	5.97	6.14	6.90		
	HDD50	3867	620	582	584	407	211	74	9	13	81	281	449	556		
	HDD65	8953	1085	1002	1049	857	658	475	330	349	487	741	899	1021		
	CDD50	394	0	0	0	0	17	50	147	131	44	5	0	0		
	CDD65	5	0	0	0	0	0	0	3	2	0	0	0	0		
	CDH74	26	0	0	0	0	0	4	16	6	0	0	0	0		
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0		
	WSAvg	7.7	7.8	7.9	8.0	7.1	8.0	8.9	8.3	7.4	6.8	6.8	7.2	7.7		
	PrecAvg	59.1	5.7	5.5	5.4	3.7	2.7	2.8	3.2	3.7	5.7	7.7	6.3	6.9		
(15) Precipitation	PrecMax	74.3	11.7	9.4	9.7	7.2	5.4	5.8	7.5	9.0	12.2	14.6	11.2	12.2		
	PrecMin	48.1	0.2	3.0	0.9	1.1	0.6	0.8	0.9	1.6	2.1	2.0	1.3	0.0		
	PrecStd	7.6	2.8	1.8	2.4	1.4	1.3	1.3	1.7	1.6	2.6	3.1	3.0	3.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	44.0	45.8	55.9	67.4	72.0	75.3	72.9	64.6	57.6	51.9	47.0		
		MCWB	41.1	40.8	38.8	46.1	52.6	56.0	61.6	59.5	54.2	50.3	46.7	42.9		
	2%	DB	43.1	41.9	43.0	50.4	60.5	63.9	68.9	67.8	61.1	54.5	48.5	45.0		
		MCWB	39.4	38.7	38.5	43.6	50.4	53.7	58.7	58.0	53.4	49.1	44.1	41.2		
	5%	DB	41.2	40.4	41.3	46.7	55.2	60.1	64.8	63.4	58.1	51.6	46.2	43.5		
		MCWB	38.0	37.6	37.5	41.5	47.7	52.4	57.6	56.7	52.4	47.6	42.4	39.7		
	10%	DB	39.2	38.7	39.8	44.2	51.8	56.5	61.2	60.6	55.8	49.8	43.9	41.6		
		MCWB	35.9	35.9	36.4	39.9	45.6	51.1	56.3	56.0	51.3	46.4	40.8	38.0		
	0.4%	WB	42.9	41.3	41.4	47.4	54.5	58.3	64.1	63.2	57.0	52.1	47.7	44.4		
		MCDB	44.4	42.6	43.1	53.5	64.5	66.7	70.4	66.8	61.1	55.6	49.4	45.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.3	39.6	39.9	44.7	51.7	55.5	60.3	59.9	54.9	50.3	45.5	42.5		
		MCDB	42.2	41.1	42.0	49.1	58.6	61.8	65.7	64.2	58.2	52.9	47.5	44.4		
	5%	WB	38.5	38.2	38.3	42.4	48.7	53.5	58.4	58.0	53.6	48.6	43.1	40.4		
		MCDB	40.7	40.0	40.7	45.9	53.7	57.9	62.9	62.2	56.6	51.1	45.4	42.9		
	10%	WB	36.3	36.0	36.6	40.5	46.7	51.8	56.9	56.4	52.2	46.7	40.9	38.4		
		MCDB	38.8	38.0	39.3	43.6	50.9	55.8	60.9	60.1	55.2	49.5	43.8	41.0		
	5% DB	MDBR	7.2	7.3	7.9	8.5	8.7	7.1	7.6	7.8	8.2	6.8	6.7	6.9		
		MCDBR	7.7	7.4	6.8	11.4	15.2	13.9	14.9	14.0	11.4	8.1	8.4	7.7		
		MCWBR	6.4	6.3	5.4	6.7	7.3	6.3	6.8	6.9	6.4	6.2	6.7	6.8		
		MCWBR	7.6	7.1	6.8	10.0	14.0	11.8	12.8	11.2	8.9	7.6	7.9	7.6		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.9	6.7	6.1	6.7	7.4	6.5	7.2	6.7	6.1	6.6	7.3	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.375	0.287	0.324	0.342	0.348	0.347	0.351	0.342	0.327	0.294	0.353	N/A		
	taud		2.546	2.115	2.212	2.341	2.439	2.485	2.504	2.518	2.511	2.244	2.522	N/A		
	Ebn at Noon		24	148	218	254	267	271	266	255	226	146	20	0		
	Edh at Noon		3	15	24	29	29	29	27	24	19	12	3	0		
(44) All-Sky Solar Radiation	RadAvg		6	95	434	973	1409	1551	1407	1017	537	177	20	0		
	RadStd		1	21	61	86	108	220	146	95	50	32	2	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.86	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page